

Asignatura	Datos del alumno	Fecha
Bases de Datos Avanzadas	Apellidos: Vargas Polania	25/05/2026
	Nombre: Javier David	

Actividades

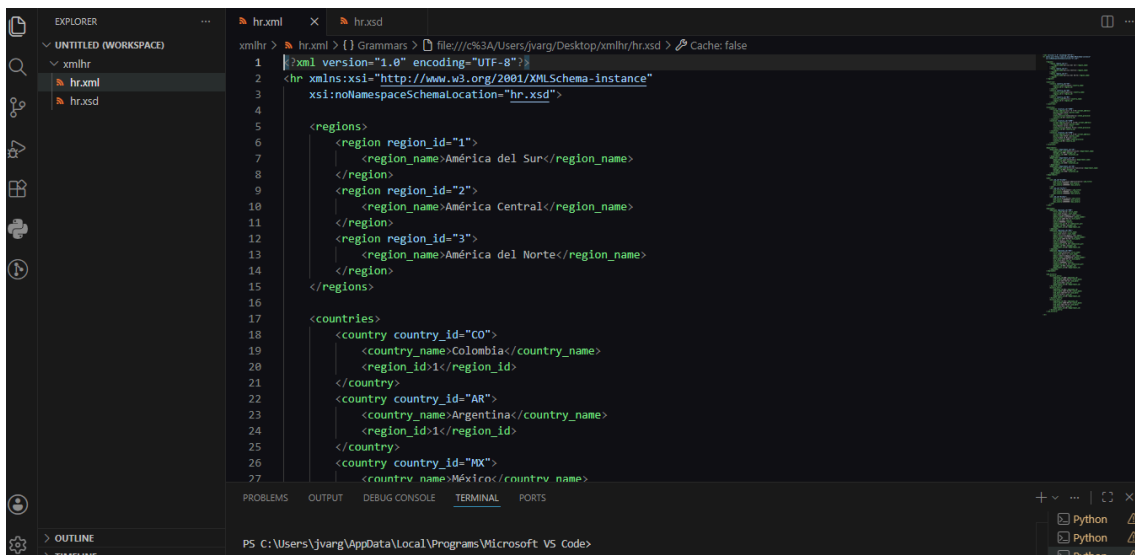
Laboratorio #1: Almacenamiento y validación de ficheros XML en Oracle

Link del repositorio

https://github.com/DavidVargasP/TareasBasesDeDatosAvanzadas2026_2.git

Primer paso

Archivo XML, que está bien formado, siguiendo las pautas.



```

1  <?xml version="1.0" encoding="UTF-8"?>
2  <hr xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3    xsi:noNamespaceSchemaLocation="hr.xsd">
4
5
6    <regions>
7      <region region_id="1">
8        <region_name>América del Sur</region_name>
9      </region>
10     <region region_id="2">
11       <region_name>América Central</region_name>
12     </region>
13     <region region_id="3">
14       <region_name>América del Norte</region_name>
15     </region>
16   </regions>
17
18   <countries>
19     <country country_id="CO">
20       <country_name>Colombia</country_name>
21       <region_id>1</region_id>
22     </country>
23     <country country_id="AR">
24       <country_name>Argentina</country_name>
25       <region_id>1</region_id>
26     </country>
27     <country country_id="MX">
28       <country_name>México</country_name>
29     </country>
30   </countries>
31 </hr>

```

Esquema XSD

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```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
3
4   <xs:simpleType name="salary_type">
5     <xs:restriction base="xs:decimal">
6       <xs:minExclusive value="0"/>
7     </xs:restriction>
8   </xs:simpleType>
9
10  <xs:simpleType name="commission_type">
11    <xs:restriction base="xs:decimal">
12      <xs:minInclusive value="0"/>
13      <xs:maxInclusive value="1"/>
14    </xs:restriction>
15  </xs:simpleType>
16
17  <xs:element name="hr">
18    <xs:complexType>
19      <xs:sequence>
20
21        <xs:element name="regions">
22          <xs:complexType>
23            <xs:sequence>
24
25              <xs:element name="region" maxOccurs="unbounded">
26                <xs:complexType>
27                  <xs:sequence>
28
29                    <xs:element name="region" type="xs:string"/>
30
31                  </xs:sequence>
32                </xs:complexType>
33              </xs:element>
34            </xs:sequence>
35          </xs:complexType>
36        </xs:element>
37      </xs:sequence>
38    </xs:complexType>
39  </xs:element>
40
41 </xs:schema>

```

Segundo paso

Debido a lo anterior, es natural decir que la validación se hace con XSD.

Tercer paso

Ahora, se muestra la asociación del XML con el XSD.

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <hr xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3   xsi:noNamespaceSchemaLocation="hr.xsd">
4
5   <regions>
6
7     <region>
8       <region type="xs:string"/>
9     </region>
10  </regions>
11 </hr>

```

Cuarto paso

Se observa que no hay errores, por lo tanto la validación es correcta.

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https://www.xmlvalidation.com/index.php?id=1&L=0

Validate an XML file

XML (eXtensible Markup Language) is a versatile format for data representation. Ensuring your XML documents are well-formed and valid is crucial for data integrity and system interoperability. Use our tool to quickly validate your XML files against standard syntax or custom schemas.

Read [here how to validate your XML files](#) (including referenced DTDs) online with just a few mouse clicks.

No errors were found

The following files have been uploaded so far:

XML document:

XML schema:

Click on any file name if you want to edit the file.



What is XML?

Quinto y sexto pasos

Se han hecho 8 consultas (5 XPath y 3 XQuery). Siempre se busca que se relacionen varias tablas.

Primera XPath

Se consultan los países que pertenecen a América del Sur. Como se puede observar, la consulta arroja a Colombia y Argentina.

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The screenshot shows the eXide IDE interface. The left sidebar displays a directory tree with 'db' as the root, containing 'apps', 'hr', 'data', 'queries', 'schema', and 'system'. The main editor area shows an XQuery file named 'Xpath1.xq' with the following code:

```
1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 $hr/countries/country[region_id = $hr/regions/region[region_name='América del Sur']/@region_id]
```

Below the code, the 'Adaptive Output' section shows the results of the query, which are two XML elements representing countries in South America:

```
1 <country country_id="CO">
  <country_name>Colombia</country_name>
  <region_id>1</region_id>
</country>
2 <country country_id="AR">
  <country_name>Argentina</country_name>
  <region_id>1</region_id>
</country>
```

Segunda XPath

Se consultan los departamentos que están en México. Como se puede observar, la consulta arroja al de Dirección Ejecutiva.

The screenshot shows the eXide IDE interface. The left sidebar displays a directory tree with 'db' as the root, containing 'apps', 'hr', 'data', 'queries', 'schema', and 'system'. The main editor area shows an XQuery file named 'Xpath2.xq' with the following code:

```
1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 $hr/departments/department[location_id = $hr/locations/location[country_id='MX']/@location_id]
```

Below the code, the 'Adaptive Output' section shows the results of the query, which is a single XML element representing the 'Dirección Ejecutiva' department:

```
1 <department department_id="30">
  <department_name>Dirección Ejecutiva</department_name>
  <manager_id>502</manager_id>
  <location_id>1700</location_id>
</department>
```

Tercera XPath

Se consultan los empleados que trabajan en Buenos Aires. Como se puede observar, la consulta arroja a Sara Conor.

The screenshot shows the eXide IDE interface. The left sidebar displays a directory tree with 'db' as the root, containing 'apps', 'hr', 'data', 'queries', 'schema', and 'system'. The main editor area shows an XQuery file named 'Xpath3.xq' with the following code:

```
1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 $hr/employees/employee[department_id = $hr/departments/department[location_id = $hr/locations/location[city='Buenos Aires']/@location_id]/@department_id]
```

Below the code, the 'Adaptive Output' section shows the results of the query, which is a single XML element representing the employee 'Sara Conor':

```
1 <employee employee_id="501">
  <first_name>Sara</first_name>
  <last_name>Conor</last_name>
  <email>saraisconor@gmail.com</email>
  <phone_number>3157634598</phone_number>
  <hire_date>2007-01-04</hire_date>
  <job_id>TecProx</job_id>
  <salary>10000000</salary>
  <commission_pct>0.16</commission_pct>
  <manager_id>502</manager_id>
  <department_id>20</department_id>
</employee>
```

Cuarta XPath

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Se consultan los empleados cuyo salario es mayor al mínimo de un programador. Como se observa, la consulta muestra a David, Sara y Catalina.

The screenshot shows the eXide IDE interface. The left sidebar displays a directory tree with 'db' as the selected folder. The main editor area shows an XQuery file named 'Xpath4.xq'. The query is as follows:

```
1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2 $hr/employees/employee[xs:decimal(salary) > xs:decimal($hr/jobs/job[job_title='Programador']/min_salary)]
```

The results pane shows three XML elements, each representing an employee:

- Employee ID 500: David Vargas, last name Vargas, email dvargas98@gmail.com, phone number 3147629045, hire date 2003-06-14, job ID AsIAdm, salary 9000000, commission 0.1, manager ID 502, department ID 10.
- Employee ID 501: Sara Conon, last name Conon, email sariconon@gmail.com, phone number 3157634598, hire date 2007-01-04, job ID TecProc, salary 10000000, commission 0.16, manager ID 502, department ID 20.
- Employee ID 502: Catalina Beltrán, last name Beltrán, email cabeltrán@gmail.com, phone number 3163985479, hire date 2008-09-14, job ID DirEje, salary 15000000, commission 0.13, manager ID 502, department ID 30.

Quinta XPath

Se consulta el trabajo del empleado cuya identificación es 502. Como se observa, es de la Dirección Ejecutiva, la presidente.

The screenshot shows the eXide IDE interface. The left sidebar displays a directory tree with 'db' as the selected folder. The main editor area shows an XQuery file named 'Xpath5.xq'. The query is as follows:

```
1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 $hr/jobs/job[@job_id = $hr/employees/employee[@employee_id='502']/job_id]
```

The results pane shows a single XML element representing a job:

```
1 <job job_id="DirEje">
  <job_title>Presidente</job_title>
  <min_salary>10000000</min_salary>
  <max_salary>20000000</max_salary>
</job>
```

Primera XQuery

Se consulta el reporte salarial de los empleados, esto incluye el nombre, el trabajo, el salario actual y los rangos del mismo.

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The screenshot shows the eXide IDE interface. On the left is a project tree with folders: db, apps, hr, data, queries, schema, and system. The 'queries' folder is expanded, showing XQuery1.xq through XQuery3.xq. The main editor displays XQuery1.xq with the following code:

```

1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 for $emp in $hr/employees/employee
4 let $job := $hr/jobs/job[@job_id = $emp/job_id]
5 return
6 <salary_report>
7   <employee>{data($emp/first_name)} {data($emp/last_name)}</employee>
8   <job_title>{data($job/job_title)}</job_title>
9   <current_salary>{data($emp/salary)}</current_salary>
10  <min_allowed>{data($job/min_salary)}</min_allowed>
11  <max_allowed>{data($job/max_salary)}</max_allowed>
12 </salary_report>

```

Below the code, the results of the query are shown. The number of results is 10. The results are displayed in a table-like format with XML tags:

```

1 <salary_report>
2   <employee>DavidVargas</employee>
3   <job_title>Asistente Administrativo</job_title>
4   <current_salary>9000000</current_salary>
5   <min_allowed>3000000</min_allowed>
6   <max_allowed>10000000</max_allowed>
7 </salary_report>
8 <salary_report>
9   <employee>SaraConor</employee>
10  <job_title>Programador</job_title>
11  <current_salary>10000000</current_salary>
12  <min_allowed>7000000</min_allowed>
13  <max_allowed>19000000</max_allowed>
14 </salary_report>
15 <salary_report>
16   <employee>CatalinaBeltrán</employee>
17   <job_title>Presidente</job_title>
18   <current_salary>15000000</current_salary>
19   <min_allowed>10000000</min_allowed>
20   <max_allowed>20000000</max_allowed>
21 </salary_report>

```

Segunda XQuery

Se consultan los empleados junto al país en el que trabajan. Como se observa, está David en Colombia, Sara en Argentina y Catalina en México.

The screenshot shows the eXide IDE interface. The main editor displays XQuery2.xq with the following code:

```

1 declare variable $hr := doc("/db/hr/data/hr.xml")/hr;
2
3 for $emp in $hr/employees/employee
4 let $dept := $hr/departments/department[@department_id = $emp/department_id]
5 let $loc := $hr/locations/location[@location_id = $dept/location_id]
6 let $country := $hr/countries/country[@country_id = $loc/country_id]
7 return
8 <employee_country>
9   <name>{data($emp/first_name)} {data($emp/last_name)}</name>
10  <country>{data($country/country_name)}</country>
11 </employee_country>

```

Below the code, the results of the query are shown. The number of results is 10. The results are displayed in a table-like format with XML tags:

```

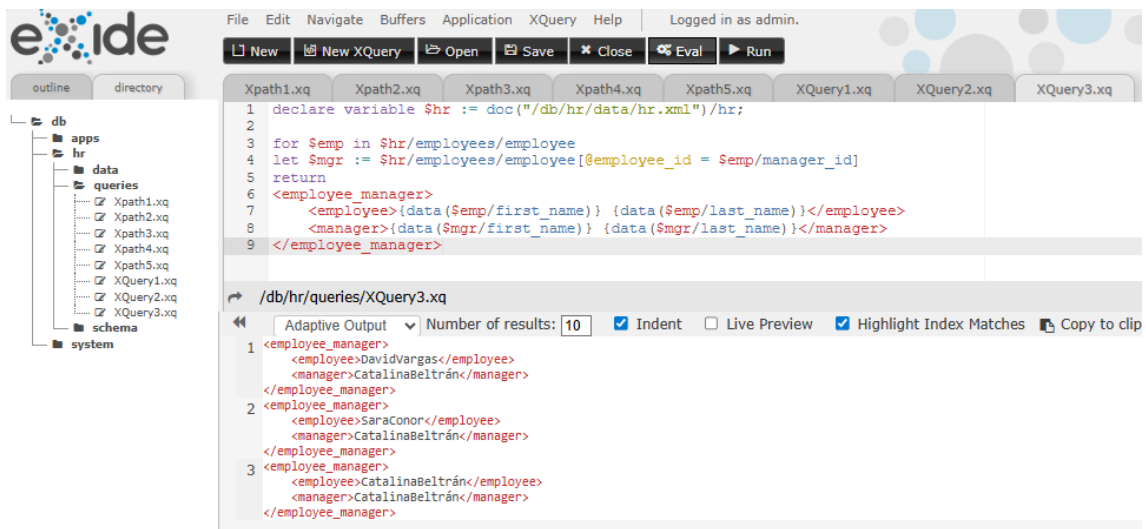
1 <employee_country>
2   <name>DavidVargas</name>
3   <country>Colombia</country>
4 </employee_country>
5 <employee_country>
6   <name>SaraConor</name>
7   <country>Argentina</country>
8 </employee_country>
9 <employee_country>
10  <name>CatalinaBeltrán</name>
11  <country>México</country>
12 </employee_country>

```

Tercera XQuery

Se consultan los empleados con su jefe inmediato. Como se observa, están los empleados con su jefe, en este caso Catalina.

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Referencias

Material del curso

GitHub - eXist-db/eXide: A web-based XQuery IDE for eXist-db · GitHub